

Multigrid Self-Gravity in the AthenaK Shearing Box: I. Mesh-Refinement Infrastructure (Phase 0)

athenak-multigrid tree, branch multigrid

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Abstract

This document opens the *multigrid track* of the self-gravity program: the route to gas + dust self-gravity in AthenaK shearing boxes *with mesh refinement*. It records Phase 0 — the shearing-box/refinement infrastructure every later phase rests on: (i) an `orbital_advection` toggle with a fully consistent non-FARGO (full-velocity) mode; (ii) an enforced *interior- x_1* refinement policy replacing AthenaK’s blanket shearing-box+refinement prohibition; and (iii) orbital advection (FARGO) on refined meshes under an *annular* refinement policy (refined regions span the full azimuthal extent). Validation: a refined-ring run reproduces a reference SMR run bit-for-bit in non-FARGO mode; with FARGO on the ring, mass is conserved exactly and a 20-orbit epicycle test tracks the analytic solution with an amplitude drift of only 0.008% (versus 0.064% on the uniform grid), with the epicyclic amplitude invariant flat to 10^{-4} . The companion document `fft_selfgravity_validation.pdf` covers the FFT solver used as the machine-precision oracle throughout this program; a Julia notebook (`multigrid_validation_tests.ipynb`) reproduces every test here.

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1 Plan and scope

1.1 Goal and roadmap

The scientific target is shearing-box simulations combining gas self-gravity, dust with mutual drag (streaming instability, including the background radial pressure gradient), and mesh refinement that zooms onto high dust-density regions in spiral arms launched by gravitational instability (cf. Baehr, Zhu & Yang 2022, ApJ 933, 100, who omitted the pressure gradient). Mesh refinement is the reason AthenaK was chosen, so the Poisson solver must ultimately live on refined meshes — which an FFT cannot. The adopted route therefore makes the *multigrid* solver shear-aware, with the validated FFT solver retained in three roles: machine-precision oracle for the multigrid shear boundary conditions, production solver for uniform-grid runs, and donor of exact open- z Dirichlet boundary values for stratified multigrid runs.

The phases of the multigrid track:

1. **Phase 0 (this document, done):** shearing box + refinement infrastructure. 0a: an `orbital_advection` on/off toggle with a consistent full-velocity (non-FARGO) mode, and the interior- x_1 refinement policy. 0b: FARGO on refined meshes under the annular refinement policy. Without 0b, refined shearing boxes pay a timestep penalty of $\sim (c_s + q\Omega L_x/2)/c_s \approx 10\text{--}30$ for gravito-turbulence boxes.
2. **Phase 1:** shear-periodic boundary conditions in the multigrid driver (`MultigridDriver`), validated against the FFT oracle: exactness at shear-periodic instants, shearing-wave convergence at generic shear phase, and the swing-amplification test with `solver=multigrid`. The V-cycle convergence rate with remapped boundaries is the one open risk of the program and is measured here.
3. **Phase 2:** multigrid gravity on refined shearing-box meshes (SMR first, then AMR), compared against high-resolution uniform FFT references.
4. **Phase 3–5:** merge with the `athenak-dust` particle-mesh dust module (dust density in the Poisson source, gravitational force on particles, pressure-gradient source term for the streaming instability), stratification with open- z boundaries, and the science runs.

1.2 Phase-0 scope and restrictions

What works after Phase 0, all enforced by fatal errors rather than conventions:

- Hydrodynamic shearing boxes with SMR, under the *uniform-boundary-level* rule: both shear-periodic x_1 boundary faces must share one refinement level. Refinement interior in x_1 always satisfies this (faces stay at root); alternatively *both* boundary faces may be refined uniformly (full x_2/x_3 extent, same level) — physically natural for a structure at the boundary, whose shear-periodic images live on both faces.
- Orbital advection on refined meshes, provided every refined region spans the full x_2 (azimuthal) extent — *annular rings* (interior, or boundary annuli as above). Without FARGO (`orbital_advection=false`) arbitrary refinement shapes are allowed subject only to the boundary-level rule.
- AMR flag handling that keeps rings intact (refine a ring if any member is flagged, derefine only if all are) — in place for Phase 0c, but AMR with a shearing box is currently fatal-guarded:

the shear-boundary GID lists (`ShearingBox::x1bndry_mbgid`) are built once at startup and go stale when AMR rennumbers MeshBlocks. Rebuilding them per AMR update is Phase 0c.

Not yet supported/exercised: AMR with a shearing box (above), MHD orbital advection with refinement (the face-centered EMF remap at level interfaces needs a $\nabla \cdot B$ analysis; fatal-guarded), and a production-size demonstration of the FARGO speedup itself. SMR/AMR runs must write `bin` outputs — the vtk writer does not support refined meshes.

2 Numerical method

2.1 Two frames for the shearing box

With background shear flow $\mathbf{v}_0 = -q\Omega x \hat{y}$, AthenaK can now evolve either of two variable sets, selected by `<shearing_box>` `orbital_advection`:

FARGO frame (`orbital_advection = true, default`). The stored velocity is the *fluctuation* $\mathbf{v}' = \mathbf{v} - \mathbf{v}_0$; the background advection is applied analytically once per step by the orbital-advection shift (Sec. 2.2). The momentum source terms are

$$\dot{M}_x = 2\Omega \rho v'_y, \quad \dot{M}_y = -(2 - q)\Omega \rho v'_x, \quad (1)$$

where the $(2 - q)$ coefficient is the sum of Coriolis (-2Ω) and the advection of background shear by the radial fluctuation ($+q\Omega$). The CFL condition sees only $|\mathbf{v}'|$.

Full-velocity frame (`orbital_advection = false`). The stored velocity includes the shear. The source terms are the full Coriolis force plus the tidal force, and (for an ideal EOS) the tidal work:

$$\dot{M}_x = 2\Omega \rho v_y + 2q\Omega^2 x \rho, \quad \dot{M}_y = -2\Omega \rho v_x, \quad \dot{E} = 2q\Omega^2 x \rho v_x \quad (2)$$

(the Coriolis force does no work). Because the wrapped data at the shear-periodic x_1 faces comes from the opposite side of the box, where the background azimuthal velocity differs by $q\Omega L_x$, the boundary exchange must offset the wrapped azimuthal momentum and energy:

$$M_y \rightarrow M_y + \rho \Delta v, \quad E \rightarrow E + M_y \Delta v + \frac{1}{2}\rho \Delta v^2, \quad \Delta v = \pm q\Omega L_x \quad (3)$$

(+ at the inner face, − at the outer). Initial conditions must include $\mathbf{v}_0$. The two frames are the same physics on different variables; comparing them is a strong consistency test (Sec. 5). The cost of the full-velocity frame is the timestep,

$$\frac{\Delta t_{\text{FARGO}}}{\Delta t_{\text{full}}} \approx \frac{c_s + q\Omega L_x/2}{c_s + |\mathbf{v}'|_{\text{max}}}, \quad (4)$$

measured directly as $5.7\times$ for the $L_x = 2\pi$ swing-test box (Sec. 5.1) and reaching $10\text{--}30\times$ for gravito-turbulence boxes — plus the extra truncation diffusion of advecting structures through the grid at many c_s . Non-FARGO is therefore the correctness baseline, not the production mode.

2.2 Orbital advection and its refinement problem

The orbital-advection step shifts every x -column of every MeshBlock by the background displacement $\delta y(x) = q\Omega x \Delta t$, as an integer number of cells plus a conservative fractional remap,

$$q_j \rightarrow q_{j-j_{\text{off}}} - (F_{j+1-j_{\text{off}}} - F_{j-j_{\text{off}}}), \quad j_{\text{off}} = \left\lfloor \frac{\delta y}{\Delta y} \right\rfloor, \quad (5)$$

with upwind remap fluxes F of first (**dc**), second (**plm**), or third (**ppmx**) order — the same kernels as the shear-periodic boundary remap. The required y -ghost data travels in buffers of depth $n_g + \text{maxjshift}$ exchanged with the two x_2 -face neighbors.

The crucial observation for refinement: *the shift kernel is already per-block correct on a multilevel mesh*. Each block computes δy from its own cell centers and applies it in its own Δy ; blocks at different levels shift by the same physical displacement expressed in their own cell units, and after the shift the ordinary AMR ghost exchange (with prolongation/restriction at x_1/x_3 level interfaces) restores consistency to interpolation order, since a uniform-in- y translation commutes with the interpolations to that order. What is *not* level-safe is the communication pattern: the buffer exchange assumes both x_2 -face neighbors are same-level, same-size blocks, which fails at any coarse-fine interface in y . This is exactly why refinement previously worked only with orbital advection disabled.

2.3 Refinement policies

Rather than generalizing the y -communication to level interfaces, Phase 0 removes the interfaces:

Uniform-boundary-level policy (always). All MeshBlocks on the two shear-periodic x_1 boundary faces must share *one* refinement level. The reason is structural: the unshifted wrap across x_1 goes through the ordinary neighbor machinery (the tree treats **shear_periodic** exactly like **periodic**), and the shear pass then redistributes the wrapped ghost data azimuthally *within each face* — with per-block Δy arithmetic and a level-aware target lookup (**FindTargetMB** shifts lx_2 modulo the block count at the source block’s own level). Nothing in that machinery references the root grid; its only real assumption is same-size blocks along each face and a same-level wrap between the faces. Hence the common level may exceed root: refining *both* boundary faces uniformly (full x_2/x_3 , same level) is legal, and a mixed-level boundary is a fatal error. Enforced block-by-block at tree build and restart (**Mesh::CheckShearingBoxRefinement**). Truly mixed-level shear boundaries (a refined patch on one face wrapping onto coarse blocks of the other) would require combining the y -shift with prolongation/restriction in the shear comms — deferred; no production code implements it.

Annular policy (when FARGO is on). Every refined region must span the full x_2 extent over its x_1 - x_3 footprint: refinement comes in *rings*. Within a ring every x_2 -face neighbor is same-level and same-size, so the per-level shift plus the *existing* communication code are valid with no changes. Enforced by ring-completeness counting over (level, lx_1 , lx_3); a partial ring with FARGO on is a fatal error naming the offending ring. For AMR, refine/derefine flags are synchronized ring-wise after the global flag exchange — refine the whole ring if any member is flagged, derefine only if every member is — deterministically on all ranks. Scientifically the ring shape is a good match for the target problem: GI spiral arms are azimuthally extended, and a ring at the arm’s radius captures the whole arm and every clump in it.

Buffer-depth safety. The integer shift on a refined block at radius x is $q\Omega|x|\Delta t/\Delta y_\ell$ cells, which the construction-time estimate `maxjshift` = $\lfloor \text{cfl} \cdot q\Omega \max|x| \rfloor + 1$ (now including the previously missing $q\Omega$ factor, doubled on multilevel meshes) may underestimate in unusual configurations. The unpack step therefore re-verifies the actual worst-case shift against the communicated buffer depth every step and aborts with a diagnostic rather than silently corrupting data.

3 What was built

Changes live in commits `e7ef37f2` (infrastructure) and `f1468da3` (bin outputs + docs); Phase 0a is ported from the `athenak` branch `test/shwave-smr`, with provenance kept in the input-file comments.

File (under <code>src/</code>)	Change
<code>hydro/hydro.cpp</code> , <code>mhd/mhd.cpp</code>	<code>orbital_advection</code> toggle; MHD+refinement fatal
<code>shearing_box/shearing_box.{hpp,cpp}</code>	store the toggle
<code>shearing_box/shearing_box_srterms.cpp</code>	dual-frame source terms, Eqs. (1)–(2)
<code>shearing_box/shearing_box_cc.cpp</code>	non-FARGO BC momentum/energy offset, Eq. (3)
<code>shearing_box/orbital_advection.cpp</code>	<code>maxjshift</code> estimate: $q\Omega$ factor, $2\times$ multilevel headroom
<code>shearing_box/orbital_advection_cc.cpp</code>	runtime shift-vs-buffer-depth guard (fatal)
<code>mesh/mesh.cpp</code>	blanket prohibition $\rightarrow$ policy warning
<code>mesh/build_tree.cpp</code>	<code>Mesh::CheckShearingBoxRefinement</code> : interior- x_1 and ring completeness
<code>mesh/mesh_refinement.{hpp,cpp}</code>	boundary-block refinement veto; ring-wise AMR flag sync
<code>pgen/tests/shwave.cpp</code> , <code>swing.cpp</code>	background shear in ICs when <code>orbital_advection=false</code>
<code>inputs/shearing_box/</code> (repo root)	<code>*_smr*.athinput</code> , <code>epicycle*.athinput</code> test inputs (bin outputs)

Runtime options (<`shearing_box`> block): `orbital_advection = true|false` (default `true`). Everything else is policy, enforced automatically. The gravity solver selection of the companion document (<`gravity`> `solver = multigrid|fft`) is unaffected by any of this; Phase 1 will add `shear_periodic` boundary support to the multigrid driver behind the same input block.

4 Building and compiling

Phase 0 needs no external dependencies beyond AthenaK’s vendored Kokkos:

```
cd athenak-multigrid
git submodule update --init kokkos      # once
mkdir build && cd build
cmake ..                                # add -D Athena_ENABLE_FFT=ON for the FFT
make -j8                                 # oracle / swing+gravity cross-checks
```

Notes: SMR/AMR runs need <`mesh`> `nghost` ≥ 4 (even, for prolongation); the FFT flag is only required for the tests that put gravity in the loop (it fetches `kokkos-fft` and needs `brew install fftw` on macOS, as detailed in the companion document). All test inputs referenced below are under `inputs/shearing_box/` and `inputs/tests/`; runs in this document were executed from `validation/run/` and `validation/run_epicycle/`.

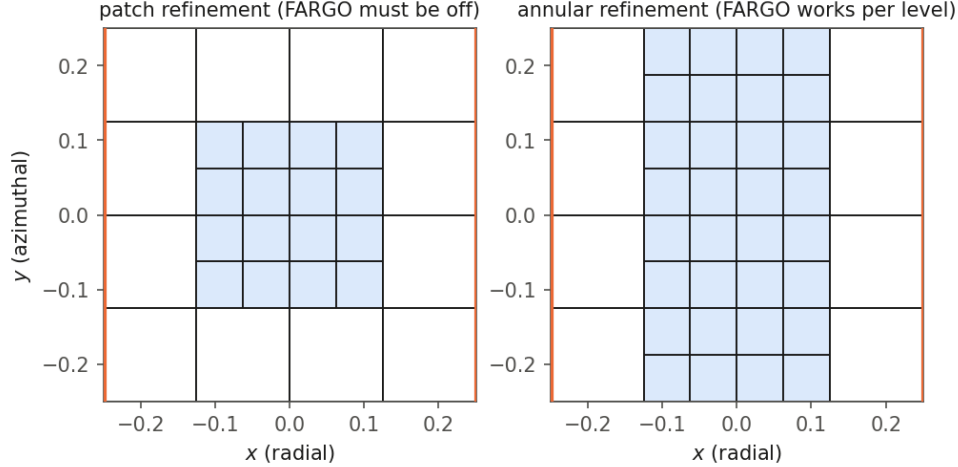


Figure 1: The two refinement layouts, drawn from the `bin` output block structure of actual runs (orange lines: shear-periodic x_1 boundaries; shaded: level-1 blocks). Left: a patch interior in both x_1 and x_2 — legal only with `orbital_advection=false`, because the coarse-fine interfaces in y break the shift communication. Right: the annular layout — the ring has no level jump in y , so FARGO runs per level with unmodified communication.

5 Tests and results

5.1 Frame consistency: FARGO vs. full-velocity, uniform grid

```
ATH=build/src/athena
$ATH -i inputs/shearing_box/hydro_incompress_shwave.athinput job/basename=shw_fargo
$ATH -i inputs/shearing_box/hydro_incompress_shwave.athinput job/basename=shw_nofargo \
    shearing_box/orbital_advection=false
# with gravity in the loop (requires FFT build):
$ATH -i inputs/tests/swing_selfgrav.athinput job/basename=SwingNoFargo \
    shearing_box/orbital_advection=false
```

The two frames agree on the frame-independent diagnostics of the JG05 vortical shearing wave — mass exactly, x -kinetic-energy history to 3.3% (the truncation gap between two different discretizations of the same problem). With self-gravity, the non-FARGO swing test tracks the linearized shearing-sheet solution with $L_2(\delta) = 1.2 \times 10^{-2}$ (FARGO: 0.9×10^{-2}) and phase-locking of the density to the instantaneous shearing wavevector at the 10^{-12} level; it costs $5.7\times$ more cycles, exactly Eq. (4).

5.2 Refinement without FARGO (Phase 0a)

```
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr.athinput \
    shearing_box/orbital_advection=false
```

An interior level-1 patch (Fig. 1, left) in the JG05 vortical-shwave problem. The run reproduces the reference run from the original `test/shwave-smr` branch **bit-for-bit** in every history column — the port changed nothing.

5.3 Guard checks (must abort)

```
# refined region snapping to a shear-boundary block: interior-x1 policy
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr.athinput \
      refined_region1/x1min=-0.24 refined_region1/x1max=-0.13
# partial-x2 ring with FARGO on: annular policy
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr.athinput \
      shearing_box/orbital_advection=true
```

Both abort with explanatory fatal errors; the second names the offending ring (“ring at level=3 lx1=2 lx3=0 has 4 of 8 MeshBlocks”).

5.4 FARGO on an annular ring (Phase 0b)

```
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_ring.athinput \
      job/basename=ring_fargo
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_ring.athinput \
      job/basename=ring_nofargo shearing_box/orbital_advection=false
```

The full- x_2 level-1 ring of Fig. 1 (right). With FARGO active on the refined mesh: mass conserved exactly (drift 0.0); x -kinetic-energy history agrees with the non-FARGO twin to 3.8% — the same truncation-level gap the two frames show on uniform grids — and with the uniform-grid FARGO run to 4.1%.

5.5 Refined x_1 boundary faces

The uniform-boundary-level rule allows refinement *at* the shear-periodic boundaries when both faces are refined uniformly. The test input refines the leftmost and rightmost root columns to level 1 over the full x_2/x_3 extent (boundary annuli), interior at root, orbital advection on:

```
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_bndry.athinput
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_bndry.athinput \
      job/basename=epi_bndry problem/ipert=1
# guard (must abort): refine only ONE face -> mixed levels across the wrap
$ATH -i inputs/shearing_box/hydro_incompress_shwave_smr_bndry.athinput \
      refined_region2/x1min=0.0 refined_region2/x1max=0.12
```

Results: the epicycle matches the analytic solution to 5.0×10^{-6} — *identical to the last digit* with the uniform-grid and interior-ring values, so the level-1 shear-periodic wrap and the level-1 azimuthal redistribution preserve uniform fields exactly. The 20-orbit endurance run on this mesh (epicycle_smr_bndry.athinput) is *bit-identical* to the interior-ring endurance run over all 202 history outputs: for a uniform field the refined shear-boundary wrap contributes exactly nothing, the strongest available statement for this configuration. The vortical shwave conserves mass exactly and agrees with the uniform run to 4.9% in x -KE (interior ring: 4.1%) and with the interior-ring run to 0.9%; a three-panel density movie of uniform | interior ring | boundary annuli (make_bin_movie_compare.jl) shows the sheared wave crossing the refined boundary with no visible artifacts. The single-face guard aborts with a message naming the level span.

5.6 The endurance test: epicycles for 20 orbits

The ipert=1 epicycle has spatially uniform $v_x(t)$, $v'_y(t)$: an exact nonlinear solution that any correct y -shift must preserve *exactly*, making it the sharpest probe of the per-level shift and its

ring-boundary communication — any indexing or buffer defect appears as $O(1)$ corruption. Two long runs ($L = 20, 128^2 \times 4$ effective resolution, ideal EOS with $c_s = \sqrt{5/3}$, $v_x(0) = 0.1 = 0.077 c_s$, 20 orbits = 20 epicyclic periods since $\kappa = \Omega$ at $q = 3/2$):

```
$ATH -i inputs/shearing_box/epicycle.athinput      # uniform 128x128x4
$ATH -i inputs/shearing_box/epicycle_smr_ring.athinput  # level-1 ring + FARGO
```

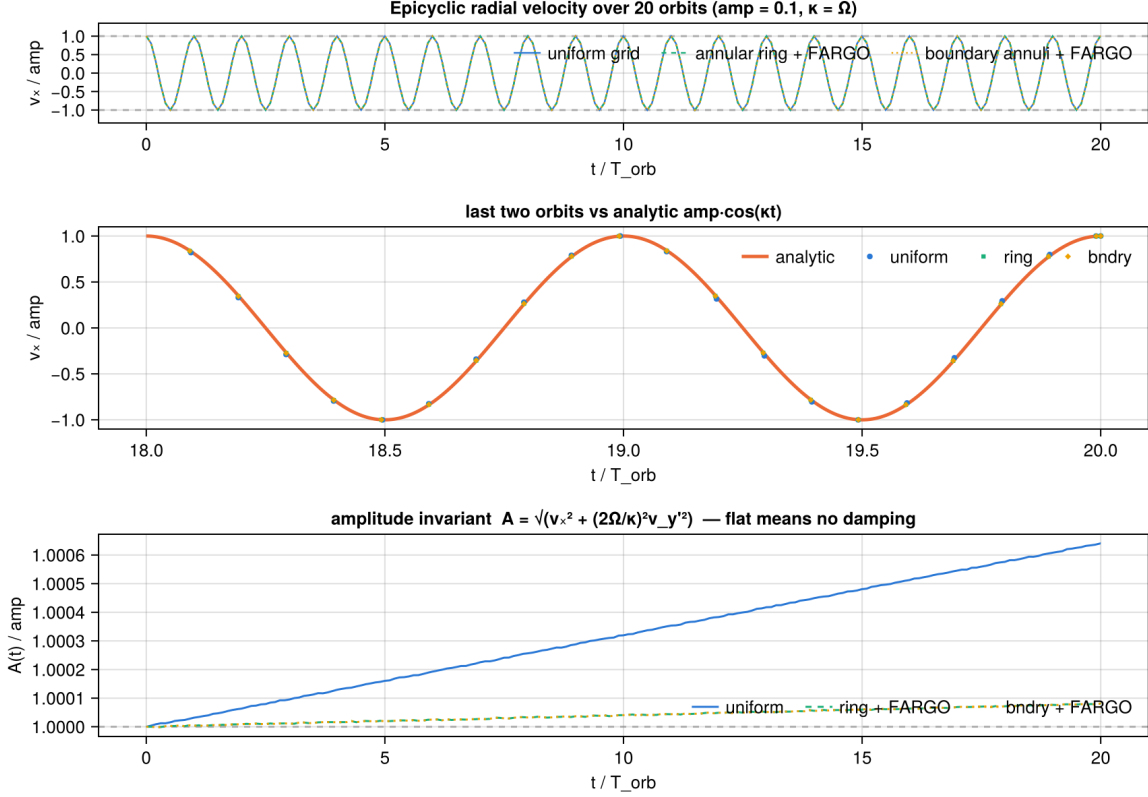


Figure 2: Volume-averaged radial velocity of the epicycle test over 20 orbits, normalized to the initial amplitude. Top: both runs oscillate between ± 1 with no visible decay. Middle: the final two orbits against the analytic $v_x = A \cos \kappa t$. Bottom: the epicyclic amplitude invariant $A = [v_x^2 + (2\Omega/\kappa)^2 v_y^2]^{1/2}$, which isolates numerical damping/growth from phase error.

run	v_x range / amp	amplitude drift, 20 orbits	max $ v_x - \text{analytic} $ / amp
uniform 128 ²	$[-1.0006, +1.0005]$	+0.064%	2.4%
annular ring + FARGO	$[-1.0000, +1.0001]$	+0.008%	0.63%
boundary annuli + FARGO	<i>bit-identical to the annular ring, all 202 outputs</i>		

Both runs hold the ± 1 envelope; the amplitude invariant is flat to a few $\times 10^{-4}$ (uniform) and 10^{-4} (ring) — a slow secular *growth* characteristic of RK2 on an oscillator, not shearing-box or refinement physics. The 3.3% peak difference between the runs is pure phase drift: the ring run takes smaller steps (finer cells) and accumulates *less* RK2 phase error, which is why the refined-mesh FARGO run is the one closer to the analytic solution. A defect in the ring communication would instead show as amplitude noise — none is present at the 10^{-5} level.

5.7 Summary

Check	Metric	Result
SMR no-FARGO vs <code>test/shwave-smr</code> reference	all history columns	bit-identical
Uniform: FARGO vs non-FARGO	mass / x -KE	exact / 3.3%
Swing + gravity, non-FARGO vs linear theory	$L_2(\delta)$ / phase	1.2×10^{-2} / 10^{-12}
Non-FARGO cost ($L_x=2\pi$ box)	cycles vs FARGO	5.7 $\times$, as predicted
Ring + FARGO: mass conservation	max drift	exact (0.0)
Shwave: ring-FARGO vs ring-non-FARGO	x -KE	3.8%
Epicycle, 20 orbits, ring + FARGO	amplitude drift / invariant	0.008% / flat to 10^{-4}
Epicycle, 20 orbits, boundary annuli	vs ring run, all columns	bit-identical (0.0)
Boundary annuli + FARGO: epicycle	max rel. vs analytic	5×10^{-6} (= uniform)
Boundary annuli + FARGO: shwave	mass / x -KE vs uniform	exact / 4.9%
Mixed-level x_1 boundary	—	fatal, names the level span
Partial- x_2 ring with FARGO	—	fatal, names the ring

Next: Phase 1 — `shear_periodic` boundary conditions in the multigrid driver, validated against the FFT oracle (exactness at shear-periodic instants; shearing-wave convergence order at generic phase; the swing test with `solver=multigrid`), with the V-cycle convergence rate under remapped boundaries as the quantity to watch.